Ryan Keith

Software Engineer · Full-Stack, Geospatial & AI Systems

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SUMMARY

Software engineer at Amgraf, Inc. since February 2024, writing code since age 14. Builds Python backend systems: data pipelines (FastAPI, Celery, PostgreSQL, AWS S3), LLM-powered tools (LangChain, Claude and OpenAI APIs) and internal IT automation (Django, Jira, Slack). Recent portfolio work adds full-stack geospatial apps: React + TypeScript frontends, FastAPI services running spatial SQL on PostGIS, and CI that tests against a real database. B.S. in Information Technology from the University of Kansas with a 4.0 GPA; Eagle Scout.

EXPERIENCE

Software Engineer · Amgraf, Inc. Kansas City, MO · Feb 2024 – Present

- Build Python backend services and data pipelines with FastAPI, Celery, PostgreSQL and AWS S3.
- Develop LLM-powered internal tools with LangChain and the Claude and OpenAI APIs.
- Automate internal IT workflows with Django, the Jira REST API and Slack integrations.

Student Technician · Johnson County Community College Overland Park, KS · Oct 2019 – Oct 2022

- Resolved student technical issues by phone and live chat, walking users through each fix step by step.
- Built a Python tool that inserts commonly sent chat replies, cutting typos and speeding up live-chat support.
- Automated a documentation ticket for every support interaction, saving technicians time after each session.
- Presented a larger application to JCCC's full IT department.

SKILLS

Languages	Python, TypeScript, JavaScript, SQL, HTML/CSS
Frameworks	FastAPI, Django, React, Vite, Pydantic
Data & GIS	PostgreSQL, PostGIS (spatial SQL, GiST indexes, KNN, hexagon grids), Redis, MapLibre GL JS, Turf.js, OpenStreetMap data
AI & ML	Claude API (tool calling), OpenAI API, LangChain, Whisper speech-to-text, sentiment analysis, satellite image change detection
Infrastructure	Docker, Celery, AWS S3, GitHub Actions CI, Render, Netlify, GitHub Pages
Testing	pytest (incl. PostGIS integration tests), Vitest, Playwright
Integrations	Stripe, SendGrid, Jira REST API, Slack webhooks, ReportLab
Professional	Technical support, user communication, leadership, planning

PORTFOLIO PROJECTS

Live apps and source code for every project: musical-marshmallow-0f8bd3.netlify.app · github.com/ryan12keith-eng/portfolio · 89 automated tests run in GitHub Actions

GeoMind AI · Smart City GIS Platform Flagship — React, TypeScript, MapLibre, FastAPI, PostGIS, Claude API

- Built a 3D map of central Kansas City with 441 real OpenStreetMap facilities and rivers, and a GIS assistant that answers plain-language questions.
- FastAPI service runs parameterized PostGIS SQL: radius search, KNN nearest facility, school-coverage gaps on a hexagon grid, density and river proximity.
- Assistant routes questions with Claude tool calling (strict schemas) when a key is configured, otherwise a rule-based router; answers always come from the database and show the SQL that ran.
- Validated inputs and rate limiting; pytest suite runs against a PostGIS container in CI; frontend falls back to in-browser Turf.js when the API sleeps.

LandScope AI · Land Suitability Scoring — React, FastAPI, PostGIS, PostgreSQL

- Measured 415 real land-use sites in a PostGIS materialized view: true area and geodesic distance to the nearest major road, school and hospital.
- Ranked sites with configurable weights (scoring rules mirrored and tested in Python and TypeScript); saved site reports in PostgreSQL with shareable links.

EarthWatch AI · Flood Change Detection TypeScript, Canvas image analysis

- Compared NASA Landsat images of the 2024 Rufiji River flood pixel by pixel to detect newly flooded land (~11% of the scene), classify land cover and compute a vegetation index.

Workflow apps · PrintFlow, Smart Quote Assistant, Job Application Tracker React, TypeScript

- Print job and proof portal with role-based permissions, an enforced status workflow, validated uploads and an audit trail; quote-email spec extraction with deterministic, unit-tested pricing; drag-and-drop application tracker with posting analysis.

EARLIER PROJECTS

Print-Job Billing & Invoicing Engine Python, FastAPI, Celery, Redis, PostgreSQL, Stripe, AWS S3

- Built an ingestion pipeline where Celery workers watch an S3 bucket and process uploaded PDF and PostScript print jobs in the background.
- Wrote a parser that extracts page count, dimensions, color profiles and vector complexity to estimate each job's cost.
- Designed a pricing API driven by a per-client pricing matrix that changes without a redeploy.
- Integrated Stripe and SendGrid to generate invoices, email clients and export accounting manifests automatically.

AI Customer Support QA Engine Python, LangChain, Claude/OpenAI API, Whisper, React, PostgreSQL

- Built a batch pipeline that ingests chat logs (JSON/CSV) and transcribes support calls with Whisper.
- Used an LLM to audit each conversation against support guidelines (greeting, identity verification, accuracy) and score technician compliance.
- Added sentiment analysis that flags rising customer frustration or wrong answers; built a React dashboard of resolution time, sentiment trends and technician scores.

IT Asset Provisioning & Inventory System Python, Django, PostgreSQL, Jira REST API, Slack, ReportLab, Docker

- Designed a schema tracking hardware serial and MAC addresses, purchase and warranty dates, and assigned employees.
- Automated onboarding so each new-hire event opens a Jira ticket with a hardware setup checklist; scheduled stock checks alert purchasing in Slack.
- Generated weekly ReportLab PDF reports on asset depreciation and assignments; containerized the app with Docker.

EDUCATION & LEADERSHIP

B.S., Information Technology · The University of Kansas Aug 2021 – May 2023 · GPA 4.0

A.S., Information Technology · Johnson County Community College 2019 – 2021

- **Eagle Scout**, Boy Scouts of America: earned at 17 after five years of leadership and service projects.
- **Full-time volunteer missionary**, British Columbia, Canada: two years of daily outreach, which built communication and self-directed planning skills.